

2002

① Finding New Plays in Old Plays: Recent Geochemical - Exploration Successes from Texas & OK: Schumacher: 065 C. 107 (2002)

- Most O&G accumulations leak, predominantly, vertically. This leakage can be detected and mapped by using any of a number of direct or indirect geochemical - exploration methods.
- Detailed geochemical & microbial surveys and research studies document that hydrocarbon microseepage from oil & gas accumulations is common and widespread, is predominantly vertical, and is dynamic. (Klusman & Saeed, 1996 & Jones & Burrill, 1996)

Microseepage Changes Over Time

- ① Park Springs Conglomerate Field, Montague Co, Texas
Wells in the heart of the conglomerate produced over a 50% reduction in microbial values in the 2nd survey (10/97) vs the 1st (2/96) and only one year after the discovery well was drilled in 10/96.
- ② SE Vassar Vertz Sand Unit in Payne County, OK
1st in 1994 and then again in 1999, 441 shallow ss. samples were collected every 100 yards, covering 1.25 m², proving there exists a dynamic microseepage system within the WF pattern. at the time of the 1st survey in 1994, the most recent & prolific wells in the No. part of the reservoir displayed positive hydrocarbon-microseepage signatures. The older wells further South showed stronger patterns in between wells in the undrained areas. The oldest abandoned wells demonstrate elevated microseepage from re-pressuring.

5/01

② Soil Gas Survey Detects Microseepage through Thick Evaporite Sequence in Egypt. El-Bishlahy: 04655-7-01

- The results of the surface exploration survey indicate clearly that the sealing capabilities of thick salt sequences are insufficient to prevent hydrocarbon microseepage to the surface from deep petroleum reservoirs.

③ Anomaly Shifts Indicate Rapid Surface Seep
Ratio: Steven Jodas: OGS, 5-29-99, p. 69-72

- Several theories have been proposed that cannot adequately explain or quantify the mechanism of hydrocarbon migration. Some authors have postulated, based on computer modeling and general calculations, that the movement is slow, taking a minimum of 10 years to go only 1000' ft. Conversely, recent unpublished work by both contractors and exploration companies suggests that the rate of migration is very rapid. *
- Soil strata consists of a dynamic system, which suggests that replenishment and migration of hydrocarbons from reservoir to the surface is much more rapid and real-time event than previously defined in the petroleum exploration literature.

Case Histories:

- Past 14 years - Surface geochemical studies over a broad range of environments - Depths of 400 to 9600 feet.
- Several long term studies with yearly repeat surveys after production established.

1) Dolley Field (6N-61W) Weld County, Colo.

- Disc 1990 Iodine Survey -
- Cret. D sand @ 6850'
- Subsequent drilling to 1/94 demonstrated close correlation to the iodine survey.
- Sampled several times over 8 year period
- A survey in early 1988 indicated most of the anomalies seen earlier were gone with production presently at stripper level.
- The Author has seen this effect over several other fields through time.

2) Keota Gas Storage Field

- 2500 Acres & covers most of mapped Anticline
- Gas is injected summer through fall for winter withdrawal.
- First sampled in summer of 1985 using the iodine method and indicated remnant anomalous values indicative of a depleted reservoir.
- Iodine & Soil Gas samples collected over a 9-month period from 7/88 to 4/89.

- Only a few minor anomalies were present in July, 1988, but sampling in Oct, 1988 showed a large anomalous population of total ethane as a percentage of total gas that reflected the injection of gas into the storage reservoir. Similar conditions existed in Jan, 1989. However, the anomalous conditions were essentially gone by April, 1989, reflecting the drawdown of the reservoir by spring due to withdrawal of gas.
- The Iodine results exhibit similar characteristics to the soil gas data. Iodine indicates a similar rapid change to the presence or absence of petroleum gases and their products in the soil.
- The absolute rate of migration is unknown, but can be construed as being very rapid.

3) Public Service Co. of Colorado's Leyden Natural Gas Storage Facility SW of Boulder in Jefferson County, Colo

- Dramatic changes in the presence of petroleum gases and iodine in the soil in relation to a gas storage facility were recently confirmed in a lawsuit by a private landowner. It was found that the Leyden Facility which stores gas in a abandoned coal mine workings at 800-1000 ft. had caused natural gas on adjacent properties by lateral migration of gas during injection and then vertical migration to the surface.

1/98

④ Geochemical Analysis of Seeps Gains Notice: Kathy Shirley: ANPB Explorer Jan, 1998. Since

- Since hydrocarbon microseepage is dynamic, seepage patterns can change rapidly in response to production-induced changes.
- "These studies have clearly demonstrated rapid variations in gas flux associated with pressure changes in the storage caverns and reservoirs."
- "Soil gas hydrocarbon concentrations above gas storage reservoirs responded within 2 to 4 days to pressure changes in the reservoir."

10/94

⑤ How iodine surveys help locate S.E. Colorado morron reservoirs? Steven Tedesco: 065 10-10-94

- Iodine is a natural integrator of hydrocarbon seepage, and it does not exhibit the short term fluctuations related to atmospheric pressure, moisture, and temperature that plague soil gas methods.

1) Jace Field:

- was discovered 7/1989. Immediately after the discovery Iodine & Soil Gas Surveys were conducted
- In 1993 a second Iodine & Soil Gas Survey was run. Both surveys indicated that the hydrocarbon seepage was diminishing as production proceeds. These "real time" changes in hydrocarbon flux have been observed by many geochemical contractors.

2) Moore-Johnson Field

- was discovered in 10/89
- Iodine survey shortly after the discovery. A 2nd survey was conducted in the summer of 1993. Both surveys correlated well with production.

3/92

⑥ An integrated interpretation for the origin of the Patrick Draw oil field seepage anomaly: Gerald K. Arps: AAPG Bull 76-3 (mar 1992) p 301-306

- a velocity for the seeping gases of 250-1000 ft/yr was calculated from the reservoir to the surface.

⑦ From megaseeps to microseeps - how geodesm can support remote gas, ... : 0968

Vertical Migration Rates

- an interesting aspect of the literature studies reveals the profound effects of vertical migration of hydrocarbons from a subsurface accumulation to the surface. These data have generally accrued from secondary recovery studies & gas storage projects where surveys taken over periods of time demonstrate hydrocarbon effects at the surface changing as a result of gas or fluid injection at depth. Some of these rates are as follows:

1) Wyoming (Patrick Draw field) 200-1000 ft/yr

- 2) West Texas: $0.6 - 2.0 \text{ m/day} = 1.8 \text{ to } 6 \text{ ft/day} = 657 \text{ to } 2190 \text{ ft/yr}$
- 3) Iowa (Keokuk Dome Gas Storage): 5 months To change surface mapping patterns at $2200 \text{ ft} \approx 5280 \text{ ft/yr}$

Luca Technologies

- Luca Technologies has recently discovered that on-going biogenic production of methane is taking place today in a number of large coal fields in the U.S. And also, contrary to current thinking, many hydrocarbon deposits such as coal beds, organic-rich shales & oil fields currently are, or have potential to produce large amounts of methane on an ongoing, real time basis. Due to naturally occurring anaerobic microorganisms,
- To date Luca has obtained 30 samples from 20 HC reservoirs throughout the US for testing.



Embargoed for release at 8:00 AM Mountain Time November 16, 2004

New Discovery by Luca Technologies of Real-time Natural Gas Formation in the Powder River Basin Offers Prospect for Renewable Energy Resource for Generations to Come

DENVER, CO (November 16, 2004): Researchers at Luca Technologies, Inc. have made a discovery regarding natural gas production in Wyoming's Powder River Basin that could lead to a renewable source of energy for generations to come. The company today announced that laboratory evidence shows that the Powder River Basin (PRB) coals are generating natural gas in real time through the ongoing activity of anaerobic microbes (bacteria that live in the absence of oxygen) resident in those coal fields. The company has termed sites where this microbial conversion of hydrocarbon deposits (coals, organic shales, or oil) to methane occurs "Geobioreactors™," and believes the careful management of such sites may offer a new long-term solution to U.S. energy needs.

Robert Pfeiffer, LUCA Technologies president and chief executive officer commented, "Our research on native coal, water and microbial samples from the PRB has determined that PRB coals can produce natural gas in real time. This finding suggests that the gas in the PRB need not be an ancient remnant of microbial activity, as generally believed, but instead is being actively created today. Moreover, we can increase or decrease methane production by PRB microbes by altering their access to water or nutrients, or halt gas production entirely by exposing the organisms to oxygen or heat sterilization. This finding holds the potential of turning what is today thought to be a finite energy resource into a renewable source of natural gas that could potentially go on for hundreds of years."

LUCA believes that in order to attempt to maximize the ultimate recovery of methane from this potentially enormous natural energy resource it will be necessary to amend certain current operating practices as well as review current legal and regulatory underpinnings of energy development. The company is currently discussing its findings with Wyoming and U.S. national agencies, as well as with major energy companies working in the PRB region.

Microbial Methane Production from Coal

It has long been known that certain ancient microorganisms are "methanogens" – microbes that generate methane by metabolizing other hydrocarbon sources. While it has also been generally accepted that much of the methane resident in coal fields was produced by such organisms, most of this production was thought to have occurred millions of years ago, when the hydrocarbon deposits were less mature and closer to the surface of the earth.

More recently, however, research has suggested that living methanogenic organisms may be present and actively forming methane within some major coalfields. LUCA scientists, employing the tools of modern biotechnology and genomics, have confirmed the presence of such microbes within anaerobic core samples from the PRB. In addition to demonstrating that methane production by these microbes can be stimulated by the introduction of additional nutrient compounds, or suppressed by heat sterilization or the introduction of oxygen, LUCA has shown that radio-labeled CO₂ (carbon dioxide) introduced to these PWB core samples is converted to radio-labeled methane. This demonstrates that the methane formation is the result of a biological process occurring today.

Sampling of U.S. Hydrocarbon Deposits and Related Biogenic Methane Potential

Name of Substrate	Amount of Substrate	Methane Conversion Potential at 100% Success Rate	Methane Generated with 1% Success Rate of Conversion	Number of Years of 23 Tcf/yr. U.S. Gas Supply at 1% Conversion
U.S. Residual Crude Oil in All States	170 Billion Barrels	680 Tcf	6.8 Tcf	0.3
Powder River Basin Coals in Montana and Wyoming (No minimum thickness cutoff)	1,300 Billion Short Tons (BST)	19,379 Tcf	194 Tcf	8.4
North Dakota Lignite	351 BST	5,232 Tcf	52 Tcf	2.3
Texas Lignite	28 BST	417 Tcf	42 Tcf	1.8
Illinois Basin Coal	365 BST	5,441 Tcf	54 Tcf	2.3
San Juan Basin Coal	358 BST	5,336 Tcf	53 Tcf	2.3
Piceance Basin Coal	289 BST	4,308 Tcf	43 Tcf	1.9
Northern Appalachia Coal	352 BST	5,247 Tcf	52 Tcf	2.3
Alaska Coal (No minimum thickness cutoff)	5,500 BST	81,984 Tcf	819 Tcf	35.6
Devonian/Ohio Shale Kerogen	444 BST	9,604 Tcf	96 Tcf	4.2
Antrim Shale Kerogen	1,300 BST	28,120 Tcf	281 Tcf	12.2
New Albany Shale Kerogen	508 BST	10,989 Tcf	110 Tcf	4.8
Total Methane Potential		177,097 Tcf	1,771 Tcf	78

-170 Billion barrels of oil is the amount of oil produced in total in the United States through 1996. Residual oil is assumed to equal the amount of produced oil and all oil is assumed to have a ratio of 2 hydrogen atoms to 1 carbon atom.

-All coal and lignite are converted assuming a 1 to 1 hydrogen to carbon ratio within those substrates. -

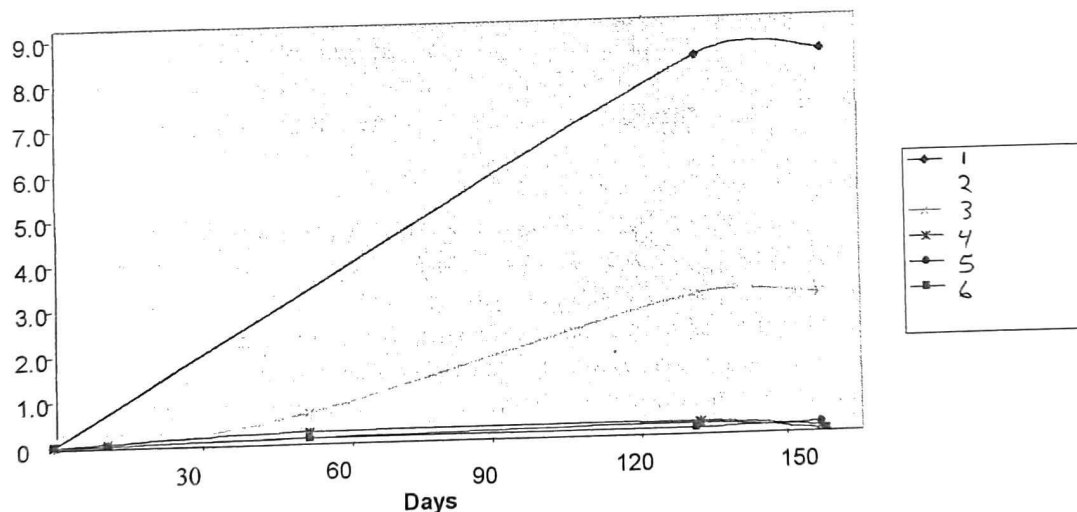
-Kerogen is assumed to have a ratio of 3 hydrogen to 2 carbon atoms.

Results & Discussion

Laboratory evidence for recent biogenic methane formation in the PRB

Luca's research demonstrates that coal seams in the PRB are living bioreactors. Unamended coal samples in formation water produced methane in real-time. Adding certain compounds to either the liquid slurry or the gas headspace can accelerate or inhibit methane production. These results are discussed below.

Gas production in unamended slurry bottles. Direct measurements of methanogenesis in carefully handled and processed coal cores from the Dietz coal seam within the PRB indicate that an active consortium of microorganisms capable of converting coal to methane exists in this coal seam. In Figure 1, methane production in coal slurries containing just formation water and crushed coal (unamended) peaked at slightly over 3% of headspace after 135 days and remained essentially level for the remainder of the experiment.



- 1- Amendment 1
- 2- Amendment 2
- 3- Unamended
- 4- Oxygen
- 5- Sterile
- 6- BESA

Figure 1. Comparison of methanogenic activities observed in Dietz coal samples under varying conditions.

While the reason for the leveling off of methane production after 135 days is not known at this time, one could surmise that nutrient limitation or inhibitor build-up may be the cause. The cessation of microbial activities is commonly observed in closed system experiments. Encouragingly, in the absence of any potentially costly amendments, methane production was significant under imperfect conditions. We have made similar findings from approximately one dozen coal samples collected from four PRB coal seams.

The sorbed gas content of PRB coals is typically in the range of 25 to 50 scf per ton of coal, generally increasing with depth. Using a correlation developed from public data for PRB coals (Crockett, 2003, personal communication), the Dietz coal seam from which this sample was taken would be estimated to contain 17 scf per ton of sorbed gas *in situ*. The rates of methanogenesis observed in these unamended slurries would generate this volume in 2.3 years. The same rates of methanogenesis would produce 50 scf per ton in 6.8 years. ✕

Methane production in amended slurry bottles. Luca's work in stimulating and inhibiting methanogenesis from PRB coals is still in its early stages. Figure 1 shows results from several experiments in which a variety of compounds were added to the slurry bottles. Amendment 2 had a modest effect above the unamended sample whereas Amendment 1 had a significant stimulatory impact on methanogenesis.

Negligible methane formation was detected in the coal slurry incubations that were sterilized, treated with BESA, or had oxygen added. Sterilization of coal samples kills all living microorganisms within the sample. BESA is a known inhibitor of methanogenesis (Muller et. al, 1993), and oxygen is a known inhibitor of methanogens (Zinder, 1993). Importantly, methanogens are considered among the most oxygen sensitive of the anaerobes and short exposures have been shown to efficiently kill several species (Zinder, 1993). Therefore it is not surprising that sterilization, BESA treatments, or exposure to oxygen will inhibit and perhaps destroy biogenesis.

Effect of water on methanogenesis. Standard experimental conditions utilized sample bottles containing 14 ml of formation water to prepare coal slurries. Table 1 identifies several types and quantities of water that were used with unamended coal samples in addition to the standard conditions. "Native" moisture samples denote slurry bottles containing coal samples without any additional water. Other experimental conditions include slurry bottles with deionized water or reduced amounts of formation water. Clearly formation water is more supportive of methanogenesis compared to deionized water. A threshold amount exceeding 0.2 ml of formation water per 5 gram sample is required for moderate methanogenesis. Two ml to fourteen ml of formation water yielded essentially identical amounts of methane.

Overcoming potential moisture and free water deficiencies in the Dietz coal is not the primary mechanism by which the addition of formation water stimulates methanogenic activity since deionized water is a poor substitute. A critical volume of formation water is necessary for biogenesis to take place and a yet to be understood interaction between coal, formation water, and the indigenous microbial consortia is required for productive methanogenesis. It should be noted that negligible methanogenic activity has been observed in experiments consisting of formation water samples without coal (data not shown).